

Are we (engineers) just optimized parts of a broken system?

Raju Doshi
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March 30, 2026

I recently reviewed the Ellen MacArthur Foundation's Butterfly Diagram and a video from *The Upshift* regarding the challenges of the circular economy.

The Diagram : https://www.ellenmacarthurfoundation.org/circular-economy-diagram Why the Circular Economy Isn't Working...
: https://www.youtube.com/watch?v=_tpDCEvD83E (*i am sharing these cause i thought it was worthwhile, nothing is sponsored)*

As a Mechatronics Engineer working with tactile intelligence, I have not given much thought to these systemic issues until now.

I have realized that my work is a direct part of the **technical cycle**, which involves the circulation of non-biodegradable materials like metals, plastics, and polymers

The video correctly highlights that creating circular products is **financially difficult** under current economic models

Traditional investment often seeks quick returns, but circular models, such as "product-as-a-service," involve longer-term cash flows that are less attractive to conventional venture capital. For these products to reach a mainstream audience we need to change the mental model of looking at higher price just as "niche luxury".

Moving forward, I am rethinking my own spending and personal habits. I plan to prioritize products designed for repair and explore sharing models for equipment that is not used daily, such as tools or machinery. As the video suggest, technical innovation must be matched by **social innovation**, such as community sharing networks, to reduce our overall material footprint

Ultimately, this makes me question our role: **should we as engineers continue to perform isolated jobs for companies that focus solely on short-term profits?** The sources suggest that circularity only works when the entire value chain collaborates, yet many companies do not even know what happens to their products at the end of their life or they choose to ignore it.

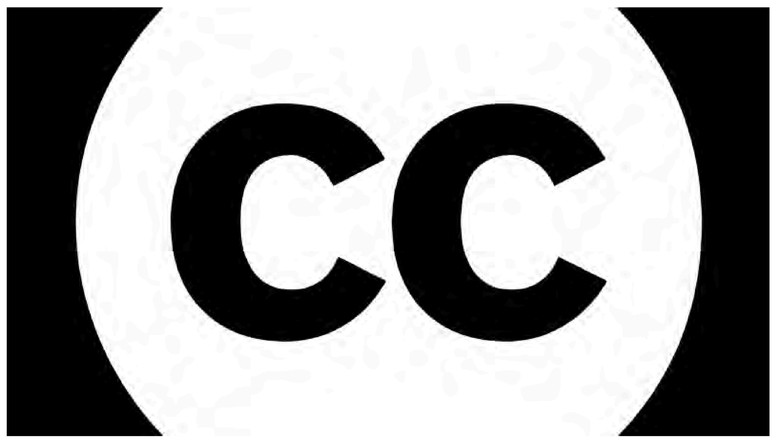
We should rethink where we work, why we work, who are we working for. It is hard to take a quick action regarding this as our livelihood depends on these jobs; but we should at least start thinking about it and be ready when the opportunity arrives.

Raju Doshi
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From "Black Box" to "Glass Box": Why Neural Mechatronics Must Be Open Source

Raju Doshi
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February 23, 2026

As a Mechatronics Engineer working with haptic neural systems, I spend my days at the literal intersection of biology and silicon. We are building machines that don't just sit on a desk—they integrate with the human body.

But as we push the boundaries of neural interfaces, we face a looming ethical "Black Box" problem.

The Trust Gap in Neural Engineering

When you buy a smartphone and it glitches, it's an inconvenience. When a neural implant—a device controlling your motor functions or interpreting your sensory input—has a proprietary "bug," it's a violation of bodily autonomy.

Right now, many neural systems are **Black Boxes**:

- The firmware is proprietary.
- The data processing algorithms are hidden.
- The user has no way to verify how their own neural data is being interpreted.

Moving to the "Glass Box" Model

If we want the public to trust neural-link technology, we must move toward a **Glass Box** approach. This is where Open Source and Creative Commons become more than just "nice-to-have" philosophies—they become safety requirements.

By utilizing open-source firmware and hardware standards, we provide three essential pillars:

1. **Auditability:** Independent researchers can verify that the code controlling a prosthetic limb is safe and won't "hallucinate" movements.
2. **Long-term Support:** If a company goes bankrupt, a user shouldn't be left with a "bricked" implant. Open-source documentation ensures that a device can be maintained by the community.
3. **Data Sovereignty:** Using Creative Commons frameworks for data sharing allows patients to own their neural patterns rather than licensing them back from a corporation.

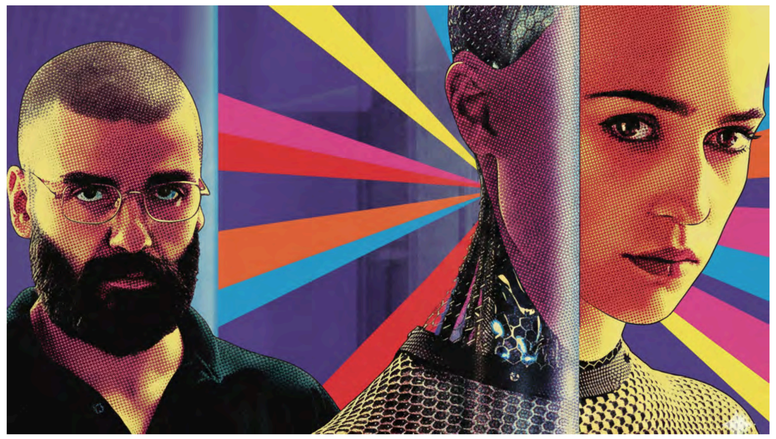
The Role of the Mechatronics Engineer

In my role, I see that Mechatronics is the bridge. We aren't just mechanical, electrical, or software engineers; we are the systems thinkers who must advocate for this transparency.

We must ask ourselves: *If I were the patient, would I want my nervous system running on a closed-source update I can't read?*

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"The Innovator's God Complex: Are we building AI to help humanity, or to prove we can play God?"

Raju Doshi
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February 22, 2026

This question keeps me up at night, especially when I think about films like 'Ex Machina.' In the movie, Nathan builds advanced AI not for societal benefit, but to satisfy his own ego and curiosity, manipulating everyone around him in the process.

The real-world parallels are striking. We're at a critical juncture where a handful of brilliant minds and powerful CEOs hold immense influence over the direction of super-intelligent AI. Their motives matter more than ever.

Is the drive primarily for safety and universal benefit, or is it a relentless race for profit, dominance, or simply the thrill of creation?

If the latter, we risk releasing models that are 'smart enough to be dangerous' but not 'safe enough to be trusted.' The allure of being first, of pushing boundaries, can sometimes overshadow the immense responsibility that comes with developing technologies that could fundamentally reshape our world.

As engineers and innovators, we must constantly interrogate our 'why.' The 'kill switch' for super-intelligence isn't just a technical feature; it's an ethical responsibility. Let's ensure our motives are aligned with human flourishing, not just unchecked ambition.

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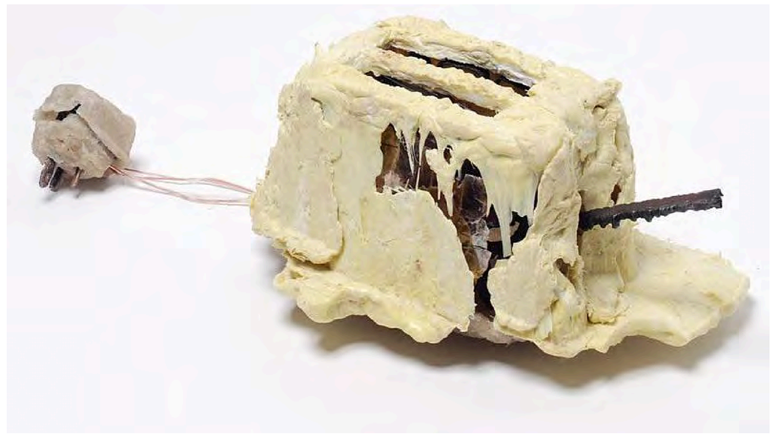
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Thomas Thwaites Toaster Project

"The Material Gap: From Thwaites' Toaster to 5,000-Mile Touch"

Raju Doshi
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February 23, 2026

Material noun: a substance that can be used for making or doing something

Matter v/s **Material** (an interesting difference) - Matter is defined by having mass and volume. Material is defined as matter with a specific atomic/molecular configuration used for a purpose.

We use lots of objects everyday, and all of them are made from some material or the other. We usually don't care to think about where these materials are coming from. And with this thought Thomas Thwaites comes into the picture. This guy took on the challenge of making a whole toaster on his own from scratch in 2009. He went to extents of mining the ore then smelting it, recreating a product of complex industrial processes. He spent nine months and 1,187 euros to make a toaster which looks like a monstrosity. Even after going to this extent he couldn't get the magic material that is 'plastic' in its raw form as he was denied access to crude oil.

He always inspires me for the efforts he put in just to understand the intricacies of creating an everyday product from scratch. Today almost 17 yrs after this project, we have moved way further from the sources of materials and we keep forgetting the hidden costs and barriers behind the source of materials.

I am currently in the research team for developing haptic feedback wearables. These devices allow an operator to feel the precise resistance and texture on their fingertips when a robot interacts with a surface, even if that surface is 5,000 miles away. The creation of almost all the advanced materials required for this technology would be unimaginable without current capabilities; or would it....? Maybe I will give it a try ;)

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Spidey sense (Spider-Man: Into the Spider-Verse 2018)

The Next Evolution: When We Can Embody More Senses

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February 23, 2026

For decades, we've viewed tools as things we hold—a hammer, a steering wheel, a smartphone. But in the world of Mechatronics, we are approaching a finish line where the tool doesn't just sit in the hand; it becomes part of the nervous system.

We are moving past the era of "using" technology and entering the era of "integrating" with it.

Extending the Five Senses

Imagine being able to "feel" the heat of a jet engine from miles away through a robotic sensor, or "seeing" in the infrared spectrum because your ocular interface translates that data directly to your brain.

We aren't just talking about fixing what is broken (like a prosthetic limb); we are talking about **sensory expansion**. Mechatronics allows us to build bridges between digital data and biological perception.

The Interface: Learning a "Sixth" Sense

The human brain is remarkably plastic; it can learn to interpret almost any data stream if delivered correctly. But how do we interface with a sense we weren't born with?

- **The Neural Handshake:** To give a human the ability to "feel" electromagnetic fields or "see" via sonar, we don't necessarily need to rewire the brain. We use Mechatronics to translate that raw data into a language the body already speaks—like vibration (haptics) or light pulses.
- **Sensory Substitution:** We can use "skin-stretching" actuators or micro-vibrations on the forearm to represent data. Over time, the brain stops thinking "my arm is vibrating" and starts thinking "there is a North-facing magnetic field to my left." The machine becomes a subconscious intuition.
- **The Seamless Link:** The goal is an interface so low-latency that the new sense feels "native." We aren't just wearing a sensor; we are expanding our biological map to include digital input.

Living in a "Hybrid" Reality

When our senses extend through Mechatronics, our definition of "self" begins to change. If you can start smelling like any animal of your choice, you will have a plethora of new information and the body as 'you' has been expanded.

This sort of modifications come with a lot of ethical and moral conundrums, but what innovation doesn't? I'm excited to see how we as humans would navigate this new world that will open up for us.

As Mechatronic engineers, our goal isn't just to build smarter robots; it's to make humans more capable of interacting with a complex world. We are the architects of the new human experience, designing the nerves and muscles of the future

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